Negotiating marine protected areas across knowledge systems: multilateral boundary work in practice

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Scholars of International Relations (IR) have understood the ocean as a space to be traversed, exploited and controlled,[1] while paying little attention to marine conservation or the negotiation of related international agreements. These increasingly oblige states to identify and manage protected areas across the whole ocean. In 2022 states adopted the Kunming–Montreal Global Biodiversity Framework under the Convention on Biological Diversity (CBD), which pursues a target of protecting 30 per cent of the ocean by 2030. In 2023 states adopted a new agreement to protect marine biodiversity in areas beyond national jurisdiction (BBNJ) under the United Nations Convention on the Law of the Sea (UNCLOS), which also provides for marine protected areas (MPAs). Defined as ocean areas managed for long-term, in-situ conservation aims, MPAs contribute to halting biodiversity loss and maintaining healthy ocean ecosystems, which mitigate climate change.[2]

However, reaching an agreement on where to establish MPAs is challenging, because it requires political consensus and a legitimate knowledge basis on which areas to protect. In a multilateral context, determining the geographical location of MPAs can be highly conflictual as it implies discussions over marine territorial divisions between states with competing interests and territorial claims, which IR scholars have explained as security and political economy issues while neglecting

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[1] Benjamin de Carvalho and Halvard Leira, 'Introduction: staring at the sea', in Benjamin de Carvalho and Halvard Leira, eds, *The sea and International Relations* (Manchester: Manchester University Press, 2022), pp. 1–25.

[2] David Johnson, Maria Adelaide Ferreira and Ellen Kenchington, 'Climate change is likely to severely limit the effectiveness of deep-sea ABMTs in the North Atlantic', *Marine Policy*, vol. 87, 2018, pp. 111–22, https://doi.org/10.1016/j.marpol.2017.09.034.

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how MPAs themselves are equally political[3] and shape maritime order. In addition to dealing with interstate conflicts over territoriality and resources, the process of multilateral MPA designation, as illustrated by the CBD, increasingly obliges states to involve non-state actors, specifically scientists and Indigenous Peoples and local communities (IPLCs). This requires new forms of cooperation and unprecedented ways of knowledge integration, specifically to bring together science and Indigenous and local knowledge (ILK) of IPLCs.[4]

Over decades, IPLCs have struggled to achieve environmental justice in biodiversity negotiations—that is, meaningful participation in decision-making, recognition for ILK and a fair distribution of the benefits and burdens of environmental protection.[5] Although the CBD formally recognizes ILK's contribution to biodiversity protection, the recognition of IPLCs' rights and epistemology is still contested within biodiversity negotiations themselves.[6] IPLCs and ILK refer to two human groups with different types of knowledge: Indigenous Peoples hold indigenous bodies of knowledge developed through generations of close contact with nature, while local communities hold local knowledge gained from close connection with specific environments, but not over several generations.[7] ILK helps define MPAs across the whole ocean because practices performed on the coast affect biodiversity in the open ocean and vice versa.[8] But ILK is often portrayed as inferior to science,[9] the latter being a knowledge system that helps define MPAs across the entire ocean and seeks to be objective by relying on research methods.[10]

We argue that the multilateral making of MPAs embraces competition and collaboration practices, inherent to what we understand as 'boundary work' spanning competing scientific and non-scientific representations of and political interests in ocean space. To illustrate that knowledge and power are inherent to boundary

[3] Philip E. Steinberg, *The social construction of the ocean* (Cambridge, UK: Cambridge University Press, 2001).

[4] Noella J. Gray, Rebecca L. Gruby and Lisa M. Campbell, 'Boundary objects and global consensus: scalar narratives of marine conservation in the Convention on Biological Diversity', *Global Environmental Politics* 14: 3, 2014, pp. 64–83, https://doi.org/10.1162/GLEP_a_00239.

[5] Kimberley R. Marion Suiseeya, 'Negotiating the Nagoya Protocol: Indigenous demands for justice', *Global Environmental Politics* 14: 3, 2014, pp. 102–24, https://doi.org/10.1162/GLEP_a_00241.

[6] Hannah Hughes and Alice B. M. Vadrot, 'Weighting the world: IPBES and the struggle over biocultural diversity', *Global Environmental Politics* 19: 2, 2019, pp. 14–37, https://doi.org/10.1162/glep_a_00503.

[7] Convention on Biological Diversity, *Training manual on the incorporation of traditional knowledge into the description and identification of EBSAs*, UNEP/CBD/SBSTTA/20/INF/21, 2016, https://www.cbd.int/doc/c/6356/dof7/5092210bd8d477491093b662/ebsa-om-2018-01-sbstta-20-inf-21-en.pdf. Indigenous Peoples are descendants from populations that have survived conquest and retained their ancestors' cultural practices. Local communities are communities that have roots in a country, might be considered a culture of the past, but have not survived conquest, such as fishermen's associations. We borrow this definition from Clement Yow Mulalap et al., 'Traditional knowledge and the BBNJ instrument', *Marine Policy*, vol. 122, 2020, https://doi.org/10.1016/j.marpol.2020.104103. (Unless otherwise noted at point of citation, all URLs cited in this article were accessible on 20 Jan. 2025.)

[8] Daniel C. Dunn et al., *Adjacency: how legal precedent, ecological connectivity, and traditional knowledge inform our understanding of proximity* (Vancouver, BC: Nippon Foundation, 2017), https://archives.nereusprogram.org/policy-brief-adjacency-how-legal-precedent-ecological-connectivity-and-traditional-knowledge-inform-our-understanding-of-proximity; Henry P. Huntington, 'Using traditional knowledge in science: methods and applications', *Ecological Applications* 10: 5, 2000, pp. 1270–4, https://doi.org/10.1890/1051-0761(2000)010[1270:UTEKIS]2.0.CO;2; Mulalap et al., 'Traditional knowledge and the BBNJ instrument'. For simplicity, we refer to Indigenous knowledge and local knowledge in the singular and as a whole—ILK—because they are both forms of knowledge different from science.

[9] Suiseeya, 'Negotiating the Nagoya Protocol'.

[10] David Johnson et al., 'The Global Ocean Biodiversity Initiative: promoting scientific support for global ocean governance', *Aquatic Conservation: Marine Freshwater Ecosystems* 29: S2, 2019, pp. 162–9, https://doi.org/10.1002/aqc.3024.

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work, we use the case of the CBD and its ecologically or biologically significant marine areas (EBSAs), which are scientific labels for marine areas within and beyond national jurisdiction that need protection. EBSA designation encourages the establishment of MPAs but does not imply enforcement measures; it promotes the use of both science and ILK.[11] Since 2012 the Conferences of the Parties (COPs) to the CBD have adopted 338 EBSA descriptions, resulting from 15 EBSA regional workshops, as well as new guidelines for modifying and identifying new EBSAs, agreed at COP16 in October 2024 after highly contested negotiations at previous COPs.[12] To analyse related boundary work, we selected four EBSA workshops: southern Indian Ocean, Arctic region, Baltic Sea and north-east Atlantic. Based on interview and text material, we identified practices that state and non-state representatives to the workshops performed to apply dissimilar epistemologies for delimiting EBSAs. Our analysis responds to recent calls to uncover how actors actually cooperate on maritime security and in ocean governance more broadly.[13]

So far, IR scholars have studied ocean politics mainly through the lenses of geopolitical realism and liberal institutionalism, and to a more limited extent through the lens of constructivism. Geopolitical realism highlights the relevance of states' location in relation to sea transportation routes and the navy's role in potential conflicts.[14] Liberal institutionalism emphasizes the legal and political challenges of ocean governance resulting from the different maritime zones under UNCLOS and the large ocean portion lying in areas beyond national jurisdiction, thus creating collective-action problems like the management of straddling fish stocks.[15] From a constructivist perspective, scientists and experts matter for ocean governance because they allow decision-makers 'to mitigate uncertainty while defending their own autonomy' by resorting to 'consensual knowledge' developed by epistemic communities.[16] These are knowledge-based groups which share common values, cause-and-effect relationships and policy-relevant conclusions, but in practice—and as our case illustrates—international environmental cooperation within the CBD includes scientists and experts from diverse backgrounds, representing different knowledge systems, world regions and values.

By focusing on the practices of creating a shared legitimate knowledge basis to identify MPAs in the CBD's context, we broaden our understanding of ocean governance and explain the effects of the diversification of 'ideas, knowledge,

[11] Convention on Biological Diversity, Decisions IX/20 and X/29.

[12] Convention on Biological Diversity, *Further work on ecologically or biologically significant marine areas: draft decision submitted by the Chair of Working Group II*, CBD/COP/16/L.8, 2024, https://www.cbd.int/doc/c/c229/9a54/cd9d051ab9a700736a73f2f4/cop-16-l-08-en.pdf; Christian Prip, 'Identifying and describing Ecologically or Biologically Significant Marine Areas (EBSAs): a key tool for the protection of ocean biodiversity in dispute', *Arctic Review on Law and Politics*, vol. 13, 2022, pp. 171–90, http://doi.org/10.23865/arctic.v13.3635.

[13] Christian Bueger, Maren Hofius and Scott Edwards, 'Global ordering and the interaction of communities of practice: a framework for analysis', *Global Studies Quarterly* 4: 1, 2024, https://doi.org/10.1093/isagsq/ksad079.

[14] Nicholas John Spykman, *The geography of the peace* (New York: Harcourt, Brace and Company, 1944); John J. Mearsheimer, *The tragedy of great power politics* (New York: W. W. Norton, 2014; first publ. in 2001).

[15] Oran Young, 'International environmental governance: building institutions in an anarchical society', in *Science and sustainability: selected papers on IIASA's 20th anniversary* (Laxenburg, Austria: International Institute for Applied Systems Analysis, 1992), pp. 245–68.

[16] Peter M. Haas, *Saving the Mediterranean: the politics of international environmental cooperation* (New York: Columbia University Press, 1990), pp. 54–5.

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and beliefs' in international environmental cooperation.[17] We adopt the perspectives of constructivism and cognitivism in IR, combining practice theories with the concepts of *boundary work* and *boundary objects* from Science and Technology Studies in order to explain, firstly, collaboration and competition in both science- and decision-making; secondly, how EBSA making relates to the steady (re)configuration of marine biodiversity governance; and thirdly, how actors determine which biodiversity knowledge is legitimate.

While previous studies of the EBSA process contributed to clarifying its (geo)politics and legal tussles, boundary-making practices and how they enable new forms of collaboration and agreement-making remain under-researched, even though this is the only multilateral effort in which state actors (scientists representing a government) identify potential MPAs together with scientists (representing non-state actors) and IPLCs. Although this new form of collaboration enabled consensus during the EBSA regional workshops, conflicts and competition arose once more when states had to endorse EBSA demarcations at CBD COPs. By first introducing the concepts of boundary work and (*composite*) boundary objects, which are empirically dealt with through a practice-driven approach, we lay the foundation for analysing the EBSA process, from the COP decision establishing regional workshops to the final endorsement of EBSAs by the COP. We describe the practices that enable collaboration and then show how the predominance of territorial interests and hierarchies between science and ILK creates competition, ultimately challenging a meaningful dialogue between states on how to use EBSAs as effective tools for marine conservation. We conclude that EBSA identification is a political act that intensifies strife among policy-makers, (re)produces science's authoritative character over ILK and reinforces IPLCs' under-representation.

Theorizing collaboration and competition

Following a constructivist and cognitivist perspective in IR, we combined two concepts, 'boundary work' and 'boundary objects', with practice theories to study the EBSA process and its outcome, EBSA descriptions. We introduce the concept of *composite* boundary objects, understood as flexible, dynamic and processual objects[18] that are themselves made up of boundary objects. Composite boundary objects are created within organized bounded spaces and are emergent because they might change boundary work practices that manage social life. They do so by constantly (re)defining boundaries across space and time.[19] This may facilitate collaboration on one site while triggering competition on another.

[17] Haas, *Saving the Mediterranean*, p. 52.

[18] Susan Leigh Star, 'This is not a boundary object: reflections on the origin of a concept', *Science, Technology, & Human Values* 35: 5, 2010, pp. 601–17, https://doi.org/10.1177/0162243910377624; Susan Leigh Star and James R. Griesemer, 'Institutional ecology, "translations" and boundary objects: amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907–39', *Social Studies of Science* 19: 3, 1989, pp. 387–420, https://doi.org/10.1177/030631289019003001.

[19] See the introduction to this special section: Maren Hofius and Matthias Kranke, 'Boundary work and the (un)making of global cooperation: mapping the terrain', *International Affairs* 101: 3, 2025, pp. 761–78, https://doi.org/10.1093/ia/iiaf061.

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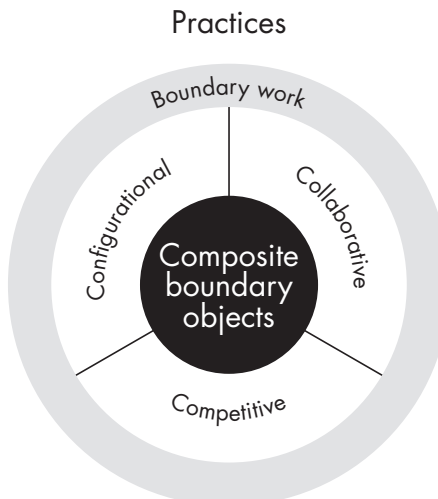
Composite boundary objects shed light on the steady (re)definition of boundaries that might lead to unexpected outcomes; and they untangle their multiple, potentially contradictory meanings. They provide a more nuanced understanding of the competition practices that are inherent to collaboration. In the following, we elaborate on boundary work, boundary objects and practice theories, which enable us to shed light on actors' continuous efforts to (re)make distinctions between human groups, marine areas and knowledge systems for the sake of power. We also elaborate on the implications for composite boundary objects.

Boundary work is the purposeful effort to (re)define boundaries and differences between groups.[20] It is part of world ordering, understood as the steady configuration of landscapes of practices that organize, stabilize and manage social life.[21] Actors battle to change the world order by transforming or creating new practices; or they seek to preserve it by preventing new practices from replacing old ones. Practices are patterns of social action that organize international affairs and are embedded with meanings that practitioners recognize. They can be configurational, collaborative or competitive (see figure 1).

Figure 1: Intertwinement of practices, boundary work and composite boundary objects



State and non-state actors performed several (sub)types of boundary work practices throughout the EBSA process (see table 1). They performed such practices to (re)define the boundaries between 1) science and ILK, 2) areas within/beyond national jurisdiction, and 3) knowledge and politics.

[20] Ann Langley et al., 'Boundary work among groups, occupations, and organizations: from cartography to process', *Academy of Management Annals* 13: 2, 2019, pp. 704–36 at p. 704, https://doi.org/10.5465/annals.2017.0089.

[21] Emanuel Adler, *World ordering: a social theory of cognitive evolution* (Cambridge, UK: Cambridge University Press, 2019), pp. 123, 141.

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Table 1: Boundary work practices in the EBSA case

Type of boundary work	*Subtype of boundary work*	*Practices*	*Actors*
Configurational (setting up bounded spaces to conduct certain activities)	Arranging	Pre-workshop training sessions	Technical support unit
	Coalescing	COP decision to adopt EBSA criteria and hold regional workshops	COP (state parties), CBD Secretariat
	Buffering	EBSA workshops	COP, CBD Secretariat
Collaborative (realigning boundaries to achieve collaboration and collective goals)	Negotiating	Creation of boundary objects (EBSA template, structured stacks of data, transponible ideas, areas coextensively inhabited by several species)	Scientists representing state or non-state actors, IPLCs, technical support unit, CBD Secretariat
	Downplaying	Avoiding the creation of methodologies that would enable science and ILK integration	Scientists representing state or non-state actors, IPLCs
Competitive (creating boundaries to obtain some advantage)	Defending	Protection and expansion boundary practices (reinforcing marine territorial divisions)	Scientists representing state actors
		Illegitimation boundary practice (expelling local communities from IPLC group)	Scientists representing state actors
	Contesting	Expansion boundary practice (contesting marine territorial divisions)	IPLCs

Sources: Ann Langley et al., 'Boundary work among groups, occupations, and organizations: from cartography to process', *Academy of Management Annals* 13: 2, 2019, pp. 704–36, https://doi.org/10.5465/annals.2017.0089; Thomas F. Gieryn, 'Boundaries of science', in Sheila Jasanoff, Gerald E. Markle, James C. Peterson and Trevor Pinch, eds, *Handbook of science and technology studies* (London: SAGE, 1995), pp. 393–443; Susan Leigh Star and James R. Griesemer, 'Institutional ecology, "translations" and boundary objects: amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907–39', *Social Studies of Science* 19: 3, 1989, pp. 387–420, https://doi.org/10.1177/030631289019003001; and authors' elaboration.

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Configurational boundary work practices

Configurational boundary work refers to outsiders' efforts to reconfigure human groups so that particular activities are conducted within bounded spaces.[22] Configurational boundary work practices strive to (re)arrange social life; they involve both competitive and collaborative elements, and have three subtypes: arranging, coalescing and buffering. These subtypes highlight the practices that allowed the EBSA regional workshops to take place. *Arranging* boundary work occurs when outsiders rearrange interactions in order to conduct new activities or old activities in a different manner. *Coalescing* happens when outsiders merge several activities in new spaces to bring together groups with conflicting interests. *Buffering* refers to creating spaces to foster collaboration between members of clashing social worlds with opposite interests.

Collaborative boundary work practices

Collaborative boundary work refers to the efforts of groups or organizations to redraw boundaries in order to enable collaboration towards collective goals.[23] Collaborative boundary work practices involve the creation of objects that facilitate collaboration,[24] and include three possible subtypes: negotiating, embodying and downplaying.[25] These subtypes unveil workshop participants' efforts to identify EBSAs collaboratively. *Negotiating* boundary work is underpinned by power relations and involves groups creating or reforming boundaries when interacting. *Embodying* refers to the activities of people that belong to, and navigate, separate social worlds.[26] *Downplaying* is what happens when members avoid debating their differences and focus on a common task.

Scholars developed this understanding of boundary work by creating the concept of boundary objects in the context of scientific knowledge production. Boundary objects explain how scientists and non-scientists cooperate even when consensus is lacking. They are material and processual work arrangements that are 'plastic enough to adapt to local needs and the constraints of the several parties employing them … They have different meanings in different social worlds' yet have a common recognizable structure.[27] Boundary objects create a common boundary that connects separate worlds, thus developing and maintaining coherence across worlds.

There are four non-exhaustive types of boundary object: repository, ideal type, coincident boundaries and standardized forms.[28] These types highlight

[22] Langley et al., 'Boundary work among groups, occupations, and organizations', p. 707.
[23] Langley et al., 'Boundary work among groups, occupations, and organizations'.
[24] Adler, *World ordering*, p. 226.
[25] Langley et al., 'Boundary work among groups, occupations, and organizations'.
[26] Table 1 lacks reference to embodying boundary practices because IPLCs and social scientists have not developed a sociocultural criterion for the identification of EBSAs that would promote the meaningful use of ILK and fill gaps in scientific data.
[27] Star, 'This is not a boundary object'; Star and Griesemer, 'Institutional ecology, "translations" and boundary objects', p. 393.
[28] Star and Griesemer, 'Institutional ecology, "translations" and boundary objects'.

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the objects used for collaboration at the EBSA regional workshops. *Repositories* are multiple catalogued objects, such as libraries. *Ideal types* loosely describe the details of places or things, making them adaptable to specific locations. *Coincident boundaries* are objects that have the same boundaries but are different internally. *Standardized forms* are objects that enable communication across dispersed groups (for example, templates). The creation of boundary objects requires the teaching and enforcement of specific methods that organize the information provided by various actors. Such methods should be simple and rigorous so that they can be applied by all and produce information that can be analysed by professionals.

Competitive boundary work practices

Competitive boundary work refers to the efforts of group members to create or defend boundaries that differentiate them from others for their own advantage.[29] Related practices seek to protect actors' divergent interests and include three possible subtypes: defending, contesting and creating. These subtypes shed light on the challenges to identify EBSAs and protect marine biodiversity. *Defending* boundary work occurs when established groups either defend or extend their domains by portraying themselves as superior and enhancing their claims. *Contesting* happens when the status of competing groups influences the tactics they employ; thus, lower-status groups might explicitly make claims about their own expertise. *Creating* refers to the attempts of weaker groups to establish their own spaces in order to portray themselves as valuable among dominant groups, or as opponents of such groups.[30]

Scholars developed the concept of competitive boundary work to explain scientists' strategies designed to differentiate science from other knowledge areas. They defined it as scientists' self-interested attempts to distinguish between what is *inside* and *outside* science.[31] Such attempts are ambiguous, flexible and context-dependent. They help scientists secure professional autonomy, deprive non-science of authority or control key issues by defining them as science.[32] Competitive boundary work practices also help demarcate scientific disciplines from each other, science from ILK and science from policy; and they prevent non-scientists from defying scientific claims and secure the political acceptability of science.

There are four non-exhaustive types of competitive boundary work: monopolization, expansion, expulsion and protection.[33] These types highlight actors' efforts to protect their interests at the EBSA regional workshops. *Monopolization*

[29] Langley et al., 'Boundary work among groups, occupations, and organizations'.

[30] Table 1 lacks reference to creating boundary practices because IPLCs have not led an EBSA workshop in which only IPLCs would participate for their meaningful involvement in the EBSA process.

[31] Thomas F. Gieryn, 'Boundary-work and the demarcation of science from non-science: strains and interests in professional ideologies of scientists', *American Sociological Review* 48: 6, 1983, pp. 781–95, https://doi.org/10.2307/2095325; Thomas F. Gieryn, 'Boundaries of science', in Sheila Jasanoff, Gerald E. Markle, James C. Peterson and Trevor Pinch, eds, *Handbook of science and technology studies* (London: SAGE, 1995), pp. 393–443.

[32] Sheila Jasanoff, *The fifth branch: science advisers as policymakers* (Cambridge, MA: Harvard University Press, 1994).

[33] Gieryn, 'Boundaries of science'.

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refers to scientists' attempts to endow an intellectual landscape with authority and credibility, while denying such attributes to other areas of knowledge. *Expansion* happens when scientists expand their cultural authority into the spaces of others or draw authority for science by linking it to the work of previous competitive boundary workers. *Expulsion* occurs when scientists 'expel not-real members from their midst' because such members illegitimately exploit science's authority.[34] *Protection* means scientists preventing outsiders, for instance policy-makers, from controlling scientific knowledge production in order to defend their resources and autonomy.

Joint configurational, collaborative and competitive boundary work practices

At the intersection between all boundary work practices, we find *composite* boundary objects. We need to unpack the diverging meanings that different actors attribute to *composite* boundary objects to understand their potential to change social life. We must understand them in relation to their dynamic scale and work arrangements, the many boundaries that they constantly (re)define, and the sites where they are created and negotiated. To this end, we studied the interrelations between configurational, collaborative and competitive boundary work practices within the EBSA process, which provides a unique opportunity to delve into the steady (re)definition of boundaries between 1) science and ILK, 2) areas within/beyond national jurisdiction, and 3) knowledge and politics.

Studying EBSAs as sites of multilateral agreement-making and boundary work

In the EBSA case, *composite* boundary objects and boundary work practices span several multilateral decision-making sites, such as the Subsidiary Body on Scientific, Technical and Technological Advice (SBSTTA) and the COPs. As an inherently hybrid process the EBSA-making combines elements of multilateral cooperation with a 'scientific and technical exercise', which we study as agreement-making. The term agreement-making 'broadens the conceptualization of the actors, sites and processes constitutive of global environmental agreements' and highlights 'how these are shaped by, reflect and have the potential to re-make or transform the intertwined global order of social, political and economic relations'.[35] Instead of conducting an analysis at the institutional level or an analysis of practices that constitute international institutions, we focus on knowledge and power-in-place by analysing the practices emerging from the EBSA process.

We studied such practices as boundary practices, which highlight actors' efforts to demarcate EBSAs, restrict what may be considered ILK and negotiate the bound-

[34] Gieryn, 'Boundaries of science', p. 432.

[35] Hannah Hughes et al., 'Global environmental agreement-making: upping the methodological and ethical stakes of studying negotiations', *Earth System Governance*, vol. 10, 2021, p. 1, https://doi.org/10.1016/j.esg.2021.100121.

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aries of marine areas needing protection, as well as the boundaries between science and ILK. We selected the following EBSA regional workshops: southern Indian Ocean, Arctic region, Baltic Sea and north-east Atlantic, because IPLCs participated in all four, enabling us to understand the use of ILK across these regions, and they took place across the northern and southern hemispheres (see table 2).[36]

Table 2: General information about selected EBSA regional workshops

Workshop		*Southern Indian Ocean*	*Arctic*	*Baltic Sea*	*North-east Atlantic*
Date		31 July–3 Aug. 2012	3–7 March 2014	20–24 Feb. 2018	22–27 Sept. 2019
Place		Mauritius	Finland	Finland	Sweden
Total number of scientists and IPLC representatives		44	31	23	35
Number of scientists (including observers)	State actors	18	13	13	22
	Non-state actors	25	16	9	12
Number of IPLC representatives		1	2	1	1
EBSAs in areas within national jurisdiction		Yes	Yes	Yes	Yes
EBSAs in areas beyond national jurisdiction		Yes	Yes	No	Yes
EBSAs straddling areas within/beyond national jurisdiction		Yes	Yes	No	Yes

Note: We assigned epistemology according to the participants' affiliation indicated on the reports. See Convention on Biological Diversity (CBD), *Report of the Southern Indian Ocean regional workshop to facilitate the description of ecologically or biologically significant marine areas*, UNEP/CBD/RW/EBSA/SIO/1/4, 2013, https://www.cbd.int/doc/meetings/mar/ebsa-sio-01/official/ebsa-sio-01-04-en.pdf; CBD, *Report of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the Baltic Sea*, CBD/EBSA/WS/2018/1/4, 2018, https://www.cbd.int/doc/c/aa9a/bde9/eaf24f73bd471d64e8094722/ebsa-ws-2018-01-04-en.pdf; CBD, *Results of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the north-east Atlantic Ocean*, CBD/EBSA/WS/2019/1/4, 2019, https://www.cbd.int/doc/c/1aba/1915/7173eb082061a278263d2fa1/sbstta-23-l-05-en.pdf; CBD, *Report of the Arctic regional workshop to facilitate the description of ecologically or biologically significant marine areas*, UNEP/CBD/EBSA/WS/2014/1/5, 2014, https://www.cbd.int/doc/meetings/mar/ebsaws-2014-01/official/ebsaws-2014-01-05-en.pdf.

[36] Indigenous Peoples and local communities did not participate in all EBSA regional workshops, further justifying our case selection. See David E. Johnson et al., 'Reviewing the EBSA process: improving on success', *Marine Policy*, vol. 88, 2018, pp. 75–85, https://doi.org/10.1016/j.marpol.2017.11.014.

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Interview methodology

The authors conducted semi-structured interviews at the CBD COP14 in 2018, online in 2021 and 2022, and in-person at the CBD COP15 in 2022. COP14 interviewees included four experts who had either participated in one of the regional workshops or were acquainted with the EBSA process. Online interviewees included 15 experts who had attended the selected workshops. COP15 interviewees included three experts that we had interviewed at COP14 or online, and two new experts who had participated in a workshop. The sample of interviewees added up to 21 experts for a total of 24 interviews. It included two IPLC representatives, one of whom has scientific training; 18 scientists, of whom eight have worked with IPLCs or used ILK; one international organization representative; and one state official.

We complemented the interviews with reports of the selected EBSA regional workshops and scientific expert workshops. We also analysed data compilations provided by the workshops' technical support unit, state and non-state actors' pre-workshop submissions, COP decisions and recommendations by the SBSTTA.[37]

EBSA-making: a combination of configurational, collaborative and competitive boundary practices

Delimiting EBSAs is a non-linear process: firstly, the COP and CBD Secretariat arranged the EBSA process so as to provide common ground for workshop participants (*configurational boundary practices*). Secondly, participants realigned the boundaries of science and ILK by creating objects that facilitated collaboration (*collaborative boundary practices*: the objects created were the EBSA template, structured stacks of data, areas coextensively inhabited by several species and transponible ideas). Finally, participants defended or contested pre-existing limits, such as marine territorial divisions (*competitive boundary practices*). In the following, we describe and analyse the three sets of boundary practices underlying the EBSA-making, basing our analysis on the four regional workshops sampled.

Configurational boundary work: setting the stage for EBSA regional workshops

The EBSA process started with discussions between state parties to the CBD on whether the convention should be mandated to identify and designate MPAs *beyond* national jurisdiction to reach its conservation objectives. A decision-making process beyond COP was needed (*bounded spaces*), where state actors could discuss, and reach a consensus on, which parts of the ocean to put aside for conserva-

[37] We did not analyse the outcomes of the 'Online Discussion Forum on Ecologically or Biologically Significant Marine Areas (EBSAs)' on the modification and identification of new EBSAs (see https://www.cbd.int/marine/forum/sept-2022?threadid=2316) because they were not reported to the COP15. Furthermore, the forum was not a workshop, nor it was requested by a COP decision.

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tion. Establishing such a particular multilateral setting requires a political mandate and COP decisions to bring together groups with conflicting interests (*coalescing*). During COPs, states had different views on whether and how the CBD should engage in EBSA-making. States needed a COP decision and (new) political mandate to define this activity's scope, mainly to avoid undermining UNCLOS regulations on the open ocean, and national sovereignty in territorial waters and exclusive economic zones. States agreed on the format of EBSA regional workshops to implement a purely 'scientific and technical exercise', creating a new multilateral process and meeting space to foster collaboration between states with opposite interests (*buffering*). To orchestrate the EBSA process, the CBD Secretariat and the technical support unit organized training workshops *rearranging* interactions between different types of actors (governments, scientists and IPLCs).

Coalescing The CBD has historically prioritized the protection of terrestrial biodiversity. This changed in 2008, when COP9 adopted the criteria to identify EBSAs in waters *beyond* national jurisdiction—that is, against dominant interpretations of the CBD's mandate that circumscribe it to areas *within* national jurisdiction—hence raising the relevance of global marine biodiversity protection within the CBD.[38]

EBSAs are 'oceanographically discrete areas that provide important services to one or more species/populations of an ecosystem or to the ecosystem as a whole'.[39] Their identification criteria are: uniqueness or rarity; special importance for species; importance for threatened or endangered species and/or habitats; vulnerability or fragility; biological productivity; biological diversity; and naturalness. COP9 also adopted scientific guidance to establish a representative MPA network. It encouraged the use of ILK to fulfil this task, and the development of biogeographic classification systems to delineate ocean regions.

In 2010, COP10 adopted EBSA identification as a 'scientific and technical exercise' that lacks enforcement measures, thus responding to persistent struggles among states over the CBD's (lack of) authority in the open ocean and the scientific or political character of EBSA identification, argued by global North and global South states correspondingly.[40] COP10 requested the CBD Secretariat to organize regional workshops. The secretariat conducted 15 workshops between 2012 and 2019, lasting one week each and focused on one specific region (see figure 2); it also published training manuals on applying the EBSA criteria and integrating ILK.[41]

[38] See Prip, 'Identifying and describing EBSAs'.

[39] Convention on Biological Diversity, Decision IX/20.

[40] Such struggles date back to the EBSA process's preparatory meetings in 2005: Prip, 'Identifying and describing EBSAs'; Gray et al., 'Boundary objects and global consensus'.

[41] Convention on Biological Diversity, Decisions IX/20 and X/29; CBD, *Training manual on the incorporation of traditional knowledge into the description and identification of EBSAs*; CBD, *Training manual for the description of Ecologically and Biologically Significant Areas (EBSAS) in open-ocean waters and deep-sea habitats*, UNEP/CBD/SBSTTA/16/INF/9, 2012, https://www.cbd.int/doc/meetings/mar/cbwsoi-wafr-01/other/cbwsoi-wafr-01-sbstta-16-inf-09-en.pdf.

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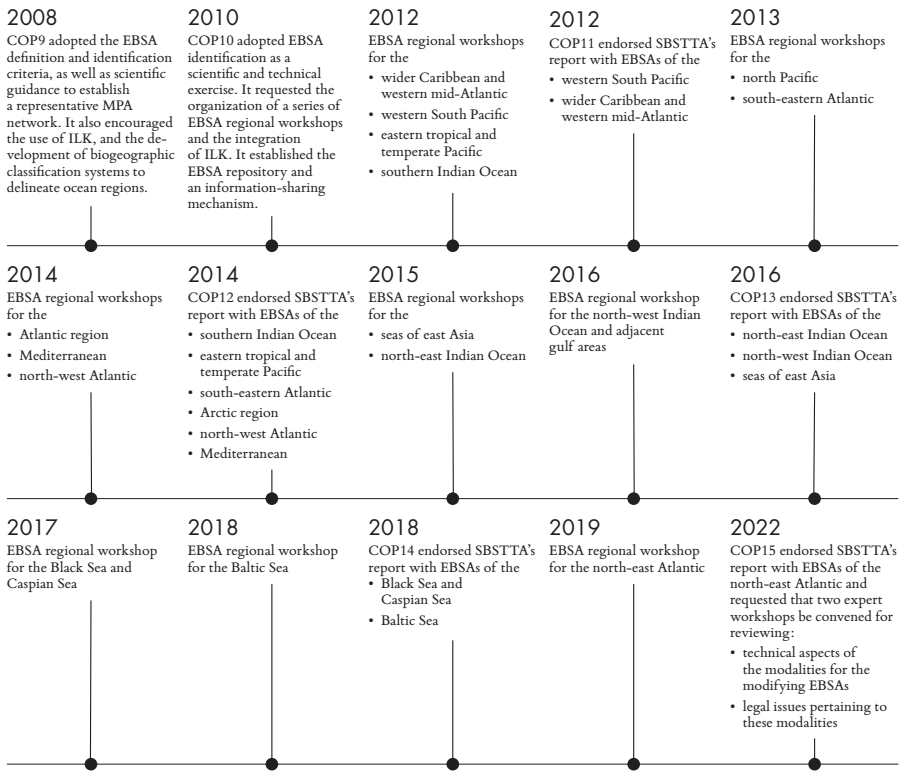
Figure 2: EBSA regional workshops and relevant COP decisions



Buffering Despite the initial focus on the open ocean, at the regional workshops states defined EBSAs in areas both within and beyond national jurisdiction (see table 2). While the secretariat invited both scientists and ILK holders to participate, states only nominated scientists—although they could have nominated ILK holders as well—thus contributing to IPLCs' under-representation at the workshops (see table 2). The technical support unit assisted participants by providing data and technical preparation. This unit consists of the Global Ocean Biodiversity Initiative, the Commonwealth Scientific and Industrial Research Organisation (Australia) and Duke University (United States).[42]

After each workshop, the secretariat sent the results to the SBSTTA, which prepared reports that specified the areas meeting EBSA criteria for endorsement by the COP.[43] After endorsement, the report was submitted to the UN General Assembly and included into the online EBSA repository (see figure 3).

[42] Personal interview, scientist, online, 9 Sept. 2021.
[43] Convention on Biological Diversity, Decision X/29.

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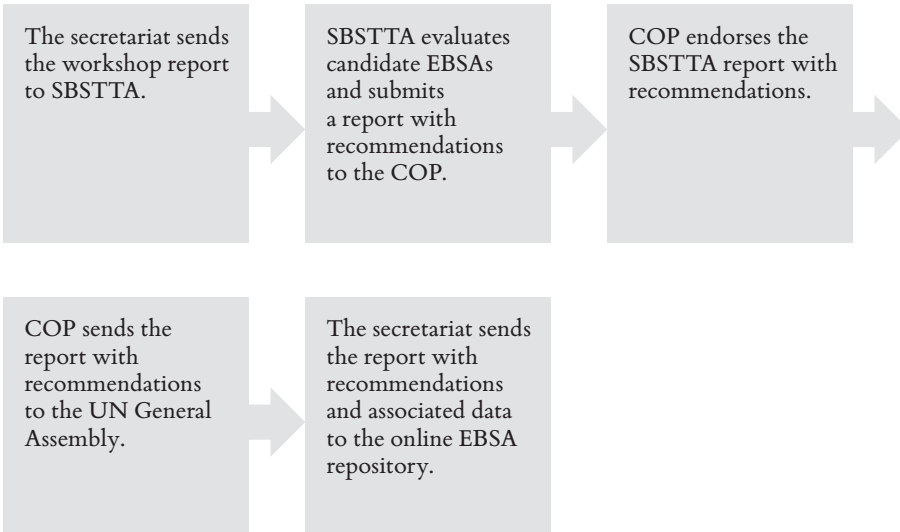
Figure 3: Flow of EBSA descriptions from regional workshops to the COP



In 2014, COP12 requested the secretariat to develop options for enhancing scientific methodologies and describing EBSAs.[44] COP13 failed to decide on such options in 2016 and requested the development of guidelines for modifying and identifying new EBSAs. At COP14 in 2018, states engaged in contentious discussions about such guidelines and failed to reach an agreement, as they feared the influence of other states over EBSAs located in disputed areas or areas *within* and *across* national jurisdictions.[45] States postponed the negotiation of these guidelines to COP15 in 2022 and then to COP16 in 2024, when states agreed on the modification and identification of new EBSAs,[46] bringing the EBSA process back in motion.

Arranging Despite states' struggles at COPs, the technical support unit complemented the CBD Secretariat's efforts by arranging boundaries in several steps. Firstly, it taught workshop participants how to understand and use data displayed in a geographic information system (GIS) and assisted them with drawing EBSA boundaries digitally. Secondly, it orchestrated the demarcation of the workshop's geographic scope. Except for that focused on the Arctic region, the EBSA regional workshops in our sample were preceded by a one-day training session led by the technical support unit.[47] Workshops would start the following day and consisted of three main phases: initial plenary, breakout groups and final plenary (see figure 4).

[44] Prip, 'Identifying and describing EBSAs'.
[45] Prip, 'Identifying and describing EBSAs'.
[46] Convention on Biological Diversity, Decision 15/26; CBD, *Further work on ecologically or biologically significant marine areas*.
[47] Personal interview, scientist, online, 9 Sept. 2021; Convention on Biological Diversity, *Report of the Southern Indian Ocean regional workshop to facilitate the description of ecologically or biologically significant marine areas*; Convention on Biological Diversity, *Report of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the Baltic Sea*; Convention on Biological Diversity, *Results of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the north-east Atlantic Ocean*.

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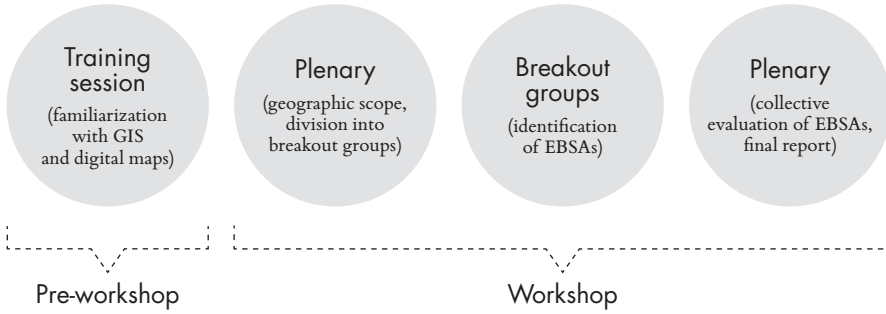
Figure 4: EBSA regional workshop phases and tasks



During the training sessions, the technical support unit described the methodology and installed an open-source GIS on participants' computers in order to provide them with digital maps and let them 'do some additional geographic information system work'.[48] These sessions brought all trainees to a minimum common level of proficiency regarding the GIS in order to improve communication.[49] They promoted particular ways of using and displaying (digital) data in order to ensure that workshop results could be processed by the technical support unit. Training sessions involved IPLCs and scientists with expertise in fisheries, the deep sea, marine mammals and genetics, among others; they enabled collaboration across these fields.[50]

During the initial plenaries, participating states determined the geographic scope by including or excluding areas within their national jurisdictions and deciding on the reach of the workshop into the open ocean. They acknowledged the sovereignty of non-participating states over certain areas by excluding them from the scope. The technical support unit facilitated these discussions by projecting the visualizations of the GIS on a screen 'with the exclusive economic zones, with the coastline' in the front of the hall, turning them on and off and asking questions such as 'Where will we draw the line between [corresponding] Island and [land]?'.[51] This constituted a simple and rigorous method to maintain coherence by ensuring that participants' efforts would focus on a common, agreed area.

Then topics for each breakout group were determined, as well as who should join each group according to their expertise.[52] This also maximized coherence and facilitated exchange between participants. Overall, the division of the ocean into a fixed geographic scope and breakout groups enabled collaboration and coherence within each workshop and across the workshop series.

[48] Personal interviews: scientist, online, 9 Sept. 2021.
[49] Star and Griesemer, 'Institutional ecology, "translations" and boundary objects'.
[50] Personal interviews: Jake Rice, Fisheries and Oceans Canada, Sharm-El-Sheikh, Egypt, 22 Nov. 2018; Jake Rice, Fisheries and Oceans Canada, Montreal, Canada, 10 Dec. 2022; scientist, online, 28 Sept. 2021; state actor, online, 20 Sept. 2021; state actor, online, 8 Sept. 2021; state actor, online, 31 Jan. 2022; scientist, online, 29 Nov. 2021; Star and Griesemer, 'Institutional ecology, "translations" and boundary objects'.
[51] Personal interview, scientist, online, 9 Sept. 2021.
[52] Personal interview, scientist, online, 9 Sept. 2021.

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Breakout groups located the coordinates of marine areas and delimited them on a paper map. The technical support unit helped with drawing 'the boundary based on the information that was there, or digitizing information that [participants]' presented,[53] such as descriptions of species and the marine environment, including 'temperature, monsoon wind [and] salinity'.[54] This technical assistance enabled the digital mapping of marine areas and had three implications. In the first place, it made it possible to define many EBSAs during a series of one-week regional workshops. Secondly, it turned digital representations into a common denominator among participants, and thus a basis for collective discussions. Thirdly, it promoted the use of digital(ized) over non-digital(ized) data, which benefited science over ILK.

During the final plenary, participants would collectively evaluate the EBSAs proposed by each breakout group. Participants would present the EBSA, and the chair would evaluate whether the data was sufficient for the area to fulfil the EBSA criteria. Then they reached consensus and classified marine areas either as EBSAs or not-EBSAs.[55]

This overall configurational boundary work set the scene ('bounded space') for states' representatives, IPLCs and scientists to engage in EBSA agreement-making. Supported by the CBD Secretariat and the technical support unit, a process was created involving four steps: 1) providing and filling out the EBSA template; 2) providing and using structured stacks of data; 3) delineating areas coextensively inhabited by several species; and 4) applying scientific criteria.

Collaborative boundary work—creating boundary objects to facilitate collaboration

By rearranging interaction between states, scientists and IPLCs, configurational boundary work created the conditions for collaborative boundary work, defined as an effort of a group or organization to redraw boundaries towards collective goals. State representatives were invited to EBSA workshops with the objective to collectively identify EBSAs in a particular region based on data and knowledge provided by the technical support unit and invited experts (including scientists and IPLCs). Each workshop resulted in a set of EBSA descriptions and maps. EBSA agreement-making was facilitated by the creation of new objects (*boundary objects*) and by *downplaying* boundary work.

EBSA workshop participants were, indeed, highly heterogeneous: state representatives from different countries, ministries and agencies; scientists with expertise in a variety of fields; and IPLCs representing multiple bodies of knowledge. While some scientists were sent to workshops as government representatives, others attended as non-state actors. *Boundary objects* were central elements supporting EBSA-making between these different interest groups and epistemologies (see figure 5).

[53] Personal interview, scientist, online, 29 Nov. 2021.
[54] Personal interview, state actor, online, 23 Feb. 2022.
[55] Personal interview, scientist, online, 28 Sept. 2021.

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Figure 5: Creating *composite* boundary objects

The *EBSA template* facilitated communication and collaboration between states, IPLCs and scientists by organizing information in a standard fashion. It required them to provide information on similar aspects of all marine areas in a consistent manner. Participants either submitted an explanation of how each marine area fulfilled the EBSA criteria ahead of the workshop or wrote 'many pages of descriptive materials' during the workshop.[56] The template also required maps, including whether the area was located within or beyond national jurisdiction, or straddled both. Then, participants rated each area's level of compliance (high, medium, low or no information) with the criteria and the scientific evidence.[57]

Structured stacks of data facilitated collaboration by providing quick access to relevant information. Participants received a report from the technical support unit containing regional and global data sets before each workshop.[58] They also received the CBD Secretariat's compilation of participants' submissions, which used science and ILK to describe EBSAs.[59] Reports and compilations were data inventories that facilitated collaboration within breakout groups, because partici-

[56] Personal interview, Tero Mustonen, ICCA (Indigenous Peoples' and Community Conserved Areas and Territories) Consortium, online, 30 Sept. 2021.

[57] Personal interview, scientist, Sharm-El-Sheikh, Egypt, 26 Nov. 2018. See the EBSA template at: https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.cbd.int%2Fdoc%2Fmeetings%2Fmar%2Febsaws-2015-01%2Fother%2Febsaws-2015-01-template-en.dot.

[58] Personal interview, scientist, online, 9 Sept. 2021.

[59] Convention on Biological Diversity, *Compilation of submissions of scientific information to describe EBSAs in the Southern Indian Ocean region*, UNEP/CBD/RW/EBSA/SIO/1/3/Rev.2, 2012, https://www.cbd.int/doc/meetings/mar/ebsa-sio-01/official/ebsa-sio-01-03-rev2-en.pdf; Convention on Biological Diversity, *Compilation of submissions of scientific information to describe areas meeting the scientific criteria for EBSAs in the Arctic region*, UNEP/CBD/EBSA/WS/2014/1/4, 2014, https://www.cbd.int/doc/meetings/mar/ebsaws-2014-01/official/ebsaws-2014-01-04-en.pdf; Convention on Biological Diversity, *Compilation of submissions of scientific information to describe areas meeting the scientific criteria for ecologically or biologically significant marine areas (EBSAs) in the Baltic Sea*, CBD/EBSA/WS/2018/1/2, 2018, https://www.cbd.int/doc/c/a2d7/2f0f/1ded49457f63a155d2dd6834/ebsa-ws-2018-01-02-en.pdf; Convention on Biological Diversity, *Compilation of submissions of scientific information to describe areas meeting the scientific criteria for ecologically or biologically significant marine areas (EBSAs) in the north-east Atlantic Ocean*, CBD/EBSA/WS/2019/1/2, 2019, https://www.cbd.int/doc/c/0e6a/7995/32829f4f873ca8ee8e0fc952/ebsa-ws-2019-01-02-en.pdf.

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pants could select and use different types of data (e.g. biological or cultural) without justifying their selection to other groups. The technical support unit also hung poster-sized versions of data from the GIS on the walls of the meeting rooms in order to provide geographical and visual information about species distribution, as well as statistical predictions of species habitats.[60] The data posters facilitated the work of breakout groups by turning the meeting room into a sort of library where participants had immediate access to diverse representations of data.

Areas coextensively inhabited by several species were identified by aggregating digital(ized) data sets concerning diverse species, thus bringing about a common space and enclosing boundaries.[61] Participants identified enclosed areas containing information and species that were relevant for all actors and by visualizing approximate calculations of species habitats on a laptop computer. They would consider one or more relevant or threatened species, determine the area where they live and draw a line around that area.[62] Participants also considered the relevance of these areas for species conservation (i.e. spawning or resting sites) in order to discard non-essential spaces from the enclosed area.[63]

Transponible ideas (such as scientific concepts) facilitated collaboration among scientists from different disciplines and fields. They commonly referred to the EBSA criterion 'connectivity of species' (agreed during *configurational boundary work*) to describe diverse marine areas and guide scientists in their joint task. They contributed to the merging of EBSAs with similar features across jurisdictions, so that transboundary EBSAs had 'ecological' instead of 'political' boundaries, making them more appealing to state representatives.[64] They partially acknowledged the ocean's connectivity and helped experts overcome fixed jurisdictional limits. In addition to ecological criteria, participants justified the protection of marine areas by describing their cultural elements or social and cultural relevance[65]—traditional uses, social acceptability or economic importance. The idea of 'social and cultural relevance' was appealing, as it can be adjusted for many areas or groups, regardless of ecological factors and data.[66]

Although the creation of those four boundary objects fostered collaboration between interest groups, it could not accommodate different epistemologies. Using both science and ILK as a basis for EBSA-making was a key CBD objective: however, methodologies on how to use ILK[67] were not provided by the secretariat.

[60] Personal interviews: scientist, online, 28 Sept. 2021; scientist, online, 9 Sept. 2021; and scientist, Sharm-El-Sheikh, Egypt, 26 Nov. 2018. Data posters can be seen in a photograph at: Global Ocean Biodiversity Initiative, 'Regional EBSA workshop for the Black Sea and Caspian Sea', https://gobi.org/regional-ebsa-workshop-for-black-sea-and-caspian-sea.

[61] Star and Griesemer, 'Institutional ecology, "translations" and boundary objects'.

[62] Personal interview, state actor, online, 8 Sept. 2021.

[63] Personal interview, scientist, online, 29 Nov. 2021.

[64] Personal interviews: state actor, online, 23 Feb. 2022; state actor, online, 11 March 2022.

[65] Convention on Biological Diversity, Decision XI/17.

[66] Star and Griesemer, 'Institutional ecology, "translations" and boundary objects'.

[67] Convention on Biological Diversity, *Report of the Arctic regional workshop to facilitate the description of ecologically or biologically significant marine areas*; Convention on Biological Diversity, *Report of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the Baltic Sea*; Convention on Biological Diversity, *Report of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the north-east Atlantic Ocean*.

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Instead of addressing and discussing the lack of methodological guidance during workshops, IPLCs and scientists performed *downplaying*, identifying EBSAs jointly without clarifying how the different knowledge systems justify the EBSA map. One participant argued that the workshop 'was not the right place to discuss about [ILK]' assuming that ILK had already 'been taken [up] during the national consultative workshop'.[68] The workshops lacked the capacity to identify the 'Indigenous knowledge attributes' of ecosystems and use Indigenous knowledge assessments of nature.[69] Simultaneously, participants used data that incorporated science and ILK, such as data concerning national fisheries and marine mammals.[70] Still, one participant had the impression that ILK 'didn't contribute at all' to delineate EBSAs.[71]

Competitive boundary work—defending and contesting marine territorial divisions

The four EBSA workshops displayed elements of competitive boundary work, protecting actors' divergent interests by creating or defending boundaries that differentiate specific groups from others for their own advantage. State actors (scientists representing a government) defended state control over areas within national jurisdiction, arguing with the value and authority of science. Boundary practices of *illegitimation*, *protection* and *expansion* performed by state actors limited IPLCs' influence on final EBSA descriptions across workshops and defended state interests in marine territory and resources. Moreover, boundary practices of *expansion* performed by IPLCs failed to increase this group's influence.

Illegitimation boundary practices to defend marine territorial divisions Building upon expulsion boundary work,[72] state actors performed illegitimation practices to expel local communities from the joint IPLCs group. The purpose was twofold: to prevent local communities from exploiting Indigenous authority to define EBSAs and to secure state scientists' professional autonomy.

At the Baltic workshop, state representatives expelled the IPLC representative—a member of a local fishermen's community—from the joint IPLCs group by not recognizing him as a legitimate member and discarding his advice.[73] They considered that, firstly, the workshop lacked Indigenous representation because the Saami people do not live in the Baltic Sea surroundings and, secondly, the participation of the local fisherman was political.[74] Additionally, state representatives only included outdated ILK in the EBSA descriptions.[75] This deprived local knowledge of its relevance and endowed science with the power to define EBSAs neutrally. It

[68] Personal interview, state actor, online, 23 Feb. 2022.
[69] Personal interview, scientist, online, 27 Oct. 2021.
[70] Personal interviews: scientist, online, 29 Nov. 2021; state actor, online, 11 March 2022; scientist, online, 27 Oct. 2021.
[71] Personal interview, Tero Mustonen, ICCA Consortium, online, 30 Sept. 2021.
[72] Gieryn, 'Boundaries of science'.
[73] Personal interviews: Tero Mustonen, ICCA Consortium, online, 30 Sept. 2021; state actor, online, 19 Oct. 2021; state actor, online, 8 Sept. 2021.
[74] Personal interview, state actor, online, 8 Sept. 2021.
[75] Personal interview, Tero Mustonen, ICCA Consortium, online, 30 Sept. 2021.

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also denied local communities the opportunity to collect and document their own knowledge while retaining ownership. Such illegitimation boundary practice is specific to the Baltic workshop because 'most areas that are important for IPLCs fall within national' jurisdiction[76] and the Baltic Sea only has these areas. Excluding all these areas from the geographic scope had brought the workshop to an end. Thus, the illegitimation boundary practice hindered IPLCs' meaningful participation in the workshop while preserving states' authority.

Protection boundary practices to defend marine territorial divisions Scientists representing states also defended state control over areas within national jurisdiction by performing protection boundary practices: they prevented workshop participants from identifying EBSAs under the jurisdiction of other states in order to preserve control over their own jurisdiction. For instance, Canada's representatives excluded areas within Canada's jurisdiction from the Arctic workshop.[77] Portugal and Spain provide an example from the north-east Atlantic workshop. These countries make competing claims over the Ilhas Selvagens/Islas Salvajes (known in English as the Savage Islands). Their experts realized that they would fail to support each other's proposals and agreed not to define any EBSAs in the disputed area.[78]

During the north-east Atlantic workshop, the geographic scope did not include any IPLC fishing areas,[79] coastal areas populated by IPLCs and areas outside IPLCs' immediate environment that are relevant for species' survival owing to ocean connectivity.[80] This practice upheld current marine territorial divisions and the authority of science over ILK in three interrelated ways. Firstly, it prevented the use of ILK advice (to identify EBSAs in areas beyond national jurisdiction) by coupling ILK applicability with areas accessed by IPLCs. Secondly, it prevented IPLCs from making authority claims over these areas. Thirdly, it provided scientists representing state actors with the autonomy required to define EBSAs in these areas. Overall, it hindered both IPLCs' meaningful participation and a fair distribution of the burdens and benefits of EBSA identification, as it did not recognize ILK's contribution to biodiversity protection.

Expansion boundary practices to contest marine territorial divisions IPLCs and state representatives conducted contesting boundary work by making explicit claims about their capacities to identify EBSAs. Specifically, they performed expansion boundary practices, which challenge the hierarchies within knowledge systems and marine territorial divisions simultaneously. They did so by including elements of one knowledge system into the territorial claim.

IPLCs provided two examples of expansion: firstly, pointing to culturally significant areas and, secondly, historical maps that they had developed. At the

[76] Johnson et al., 'Reviewing the EBSA process', p. 82.
[77] Convention on Biological Diversity, *Report of the Arctic regional workshop to facilitate the description of ecologically or biologically significant marine areas*.
[78] Personal interview, state actor, online, 11 March 2022.
[79] Personal interviews: scientist, 29 Nov. 2021; international organization representative, 1 Dec. 2021 (both online).
[80] Convention on Biological Diversity, *Report of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the north-east Atlantic Ocean*; Dunn et al., *Adjacency*.

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Arctic workshop, IPLCs argued that they depended on culturally significant areas to keep up their traditions.[81] They defined an area straddling Canada and Greenland as an EBSA on the basis of its highly productive ecosystems of species that are culturally relevant for Inuit, and its ecological properties.[82] That is, IPLCs included science *into* ILK to 1) gain cultural authority for ILK and IPLCs in policy-making, 2) introduce a cultural criterion, 3) contest state control over the Arctic and 4) gain Inuit control over the area. However, the proposed area was not designated as an EBSA because Canada and Denmark excluded waters within their national jurisdictions from the workshop's geographic scope.[83]

To facilitate the identification of EBSAs, the IPLC representative at the Baltic workshop proposed an ILK atlas that would display maps with IPLCs' intergenerational activities and ecological data on customary harvesting areas.[84] Once more, IPLCs included science *into* ILK to 1) challenge the superiority of science over ILK, 2) support IPLCs' cultural and territorial claims in policy-making and 3) contest state control over marine areas. However, the proposal for an ILK atlas was also unsuccessful.[85] Overall, IPLCs failed to achieve environmental justice by performing expansion boundary practices.

Canada provides an example of a state that performed an expansion boundary practice at the Arctic workshop. Since IPLCs have gathered more observations than scientists about Arctic marine species given their uninterrupted presence in the region over centuries, Canada mentioned the use of both science and ILK in its national EBSA process.[86] That is, Canada included ILK *into* science to challenge the superiority of ILK over science in the Arctic, and to support state authority and territorial claims over Arctic areas against those of IPLCs.

Competitive boundary work practices significantly shaped final EBSA descriptions and maps, thus manifesting state control over marine territory and hierarchies between science and ILK. In the following section, we will discuss what our analysis implies for the role that EBSA agreement-making might play as a blueprint for future international cooperation.

EBSA agreement-making: failure or blueprint for future international cooperation?

This article identified new forms of multilateral collaboration and agreement-making in ocean governance by adopting a constructivist and cognitivist perspective in IR and combining the concepts of boundary work and boundary objects with a practice-driven approach. We used the EBSA case to draw attention to

[81] Personal interview, IPLCs' representative, online, 21 Feb. 2022.
[82] Personal interview, IPLCs' representative, online, 21 Feb. 2022.
[83] Convention on Biological Diversity, *Report of the Arctic regional workshop to facilitate the description of ecologically or biologically significant marine areas*.
[84] Personal interview, Tero Mustonen, ICCA Consortium, online, 30 Sept. 2021.
[85] See Convention on Biological Diversity, *Report of the regional workshop to facilitate the description of ecologically or biologically significant marine areas in the Baltic Sea*.
[86] Convention on Biological Diversity, *Report of the Arctic regional workshop to facilitate the description of ecologically or biologically significant marine areas*; Huntington, 'Using traditional knowledge in science'.

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emerging multilateral decision-making sites in the marine conservation context characterized by an unprecedented interaction between state actors, scientists and IPLCs. To address the issue of MPAs within and beyond national jurisdiction in the CBD context, state parties supported by the CBD Secretariat and a technical support unit designed and established regional workshops targeting representatives of different knowledge systems.

To understand what this implies for the future of international cooperation and the role attributed to science and ILK in ocean governance, we analysed the EBSA process as a set of boundary work practices and objects. We differentiated between configurational, collaborative and competitive boundary work to emphasize the coexistence of collaboration and competition and found that the organizational and technical set-up of the EBSA workshops, descriptions and maps played a particular role in fostering collaboration and downplaying competing interests, values and epistemologies.

We conclude that EBSA agreement-making differs from previous forms of international cooperation, as described by epistemic communities.[87] Involved experts held widely different knowledge backgrounds, values and beliefs. In addition, the boundaries between state representatives and different knowledge holders were blurred by the multiple roles that participants played during a highly technical and data-intense process. Not alluding to the lack of methodology on how to bridge knowledge systems, and the different meanings attributed to specific marine areas under consideration, downplayed epistemological differences. However, it also created competition between participants on what counts as legitimate knowledge and how to justify new marine territorial divisions.

By detailing different sets of boundary work practices, we concluded that EBSA agreement-making is inherently political for three main reasons. Firstly, EBSAs are political because of the contested role of the CBD regarding open ocean marine biodiversity conservation, with areas *beyond* national jurisdiction falling under the mandate of UNCLOS. Secondly, EBSA descriptions in areas *within* national jurisdiction are tangent to national sovereignty, which implies that government representatives adopted a cautious approach ensuring state control regardless of the meaning of areas for ecological, cultural, social or spiritual purposes. The third reason why EBSAs are political relates to their role as *composite* boundary objects. As such, EBSA descriptions bring (un)expected, potentially contradictory outcomes. Among the expected ones, their ascribed roles are mere scientific descriptions of marine areas and valuable information for the UN General Assembly.[88] The more unexpected outcomes are that they may be policy-prescriptive and may prejudge new marine territorial divisions. Because COPs endorse EBSAs to inform CBD implementation, COPs exercise foreign influence in areas within national jurisdiction and over IPLCs. The emergent nature of EBSA descriptions is also palpable in the recent changes regarding their material and processual work

[87] Haas, *Saving the Mediterranean*, p. 52.
[88] Convention on Biological Diversity, Decision X/29.

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arrangements at workshops, signalling that they are liable to be altered again in the future.[89]

The competitive and emergent character of EBSA descriptions points to the political nature of MPAs more generally. Indeed, state efforts at COPs seek to prevent scientists and IPLCs from making international policies to protect or expand states' power over marine areas, disregarding a fair distribution of costs and benefits of biodiversity protection. Similar to what scholars observed in the contexts of maritime security and geopolitics, marine conservation and the recognition of different knowledge forms are highly political, mirroring deeply-rooted conflicts in ocean governance, such as the continuation of colonially-shaped marine science systems and the concentration of research capacity and scientific collaboration capital in the global North.[90] From this perspective, the multilateral nature of the EBSA agreement-making process not only legitimized state authority over marine territory and knowledge, but also reflected the dominant global order of social, political and economic relations.

Still, the EBSA process allows for the potential transformation of those relations. Recognizing the multiple values of nature, as exemplified by the Intergovernmental Science–Policy Platform on Biodiversity and Ecosystem Services (IPBES) values assessment, might support future EBSA workshops and increase the meaningful inclusion of ILK. According to IPBES, 'the diversity of nature's values in policymaking can be advanced by considering a typology of nature's values that encompasses the richness of people's relationships with nature'.[91] Indigenous assessments and documented or quantified ILK could support IPLCs' claims during EBSA workshops—and subsequent SBSTTA and COP negotiations. Such a process could be supported by creating new 'bounded spaces': expert workshops and local workshops focusing on knowledge integration, rather than national representation, orchestrated by social scientists and anthropologists with the support of IPBES, which has developed extensive expertise in conducting multidisciplinary environmental assessments in a multilateral setting with the ambition to foster transformative change. In the IPBES manner, transformative change implies critically reflecting and changing the decision-making processes and practices established to tackle the global environmental crises. We have attempted to do so with our analysis of the EBSA case, which displays at least some elements of a blueprint for future international cooperation.

Future research will encounter challenges when studying future EBSA workshops because of the lack of access for stakeholders and researchers beyond those invited by states and the CBD Secretariat. Researchers must rely on participants' accounts to reconstruct practices performed at workshops. Furthermore,

[89] See Convention on Biological Diversity, *Further work on ecologically or biologically significant marine areas*.

[90] See Petro Tolochko and Alice B. M. Vadrot, 'The usual suspects? Distribution of collaboration capital in marine biodiversity research', *Marine Policy*, vol. 124, 2021, https://doi.org/10.1016/j.marpol.2020.104318.

[91] Unai Pascual et al., eds, *Summary for policymakers of the methodological assessment report on the diverse values and valuation of nature of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services* (Bonn: IPBES Secretariat, 2022), p. 11.

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non-Indigenous researchers might struggle to interview IPLC representatives because researchers commonly lack training on the ethical use of ILK.

Research should focus on collaborative and competitive practices at future EBSA workshops, SBSTTA and COP meetings in order to untangle the many boundaries that such practices (re)define constantly and to elucidate injustice in marine biodiversity protection. Our practice-based study of configurational, collaborative and competitive boundary work enabled us to conceptualize EBSAs as *composite* boundary objects with the potential to change the CBD's role in marine biodiversity governance. This offered insights into challenges and avenues for the drafting of marine biodiversity protection policy at agreement-making sites, while enriching the scholarly literature about the legal and (geo)political aspects of the EBSA process.

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